

Para conectarse a ssh

```
ssh -i ~/.ssh/other20_key -p 2222 root@192.117.30.20
```

Eliminar el privilegio de sudo sin contraseña

Revisar donde esta el privilegio

```
root@c1246db0b82f: ~
Archivo Acciones Editar Vista Ayuda

(proyecto@kali-117)-[~]
$ grep -r insecureuser /etc/sudoers /etc/sudoers.d/
grep: /etc/sudoers: Permiso denegado
grep: /etc/sudoers.d/kali-grant-root: Permiso denegado
grep: /etc/sudoers.d/ospd-openvas: Permiso denegado
grep: /etc/sudoers.d/README: Permiso denegado

(proyecto@kali-117)-[~]
$ ssh root@OTHER.20
ssh: Could not resolve hostname other.20: Name or service not known

(proyecto@kali-117)-[~]
$ ssh root@192.117.30.20
root@192.117.30.20's password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 6.14.0-37-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
root@c1246db0b82f:~# mkdir -p /root/evidencias/hardening
root@c1246db0b82f:~# sudo -l -U insecureuser
Matching Defaults entries for insecureuser on c1246db0b82f:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin

User insecureuser may run the following commands on c1246db0b82f:
    (ALL : ALL) ALL
    (ALL) NOPASSWD: ALL
root@c1246db0b82f:~# grep -r insecureuser /etc/sudoers.d/
/etc/sudoers.d/insecureuser:insecureuser ALL=(ALL) NOPASSWD:ALL
root@c1246db0b82f:~#
```

Se decide removerlo por completo del grupo o de los permisos de sudo, ya que no hay justificación para que insecureuser tenga control total de root

```
(ALL) NOPASSWD: ALL
root@c1246db0b82f:~# grep -r insecureuser /etc/sudoers.d/
/etc/sudoers.d/insecureuser:insecureuser ALL=(ALL) NOPASSWD:ALL
root@c1246db0b82f:~# deluser insecureuser sudo
Removing user `insecureuser' from group `sudo' ...
Done.
root@c1246db0b82f:~# rm /etc/sudoers.d/insecureuser
root@c1246db0b82f:~# sudo -l -U insecureuser
User insecureuser is not allowed to run sudo on c1246db0b82f.
root@c1246db0b82f:~#
```

Antes: ilustración 3

Corregir permisos del directorio home de insecureuser

Ahora solo insecureuser accede a su directorio

```

root@c1246db0b82f:~# chmod 750 /home/insecureuser
root@c1246db0b82f:~# ls -la /home/insecureuser
total 36
drwxr-x--- 1 insecureuser insecureuser 4096 Sep 23 17:00 .
drwxr-xr-x 1 root          root          4096 Jul 29 06:23 ..
-rw----- 1 insecureuser insecureuser  364 Sep 23 17:00 .bash_history
-rw-r--r-- 1 insecureuser insecureuser  220 Feb 25  2020 .bash_logout
-rw-r--r-- 1 insecureuser insecureuser 3771 Feb 25  2020 .bashrc
drwx----- 2 insecureuser insecureuser 4096 Sep 23 16:33 .cache
-rw-r--r-- 1 insecureuser insecureuser  807 Feb 25  2020 .profile
-rw-r--r-- 1 insecureuser insecureuser    0 Sep 23 16:35 .sudo_as_admin_successful
drwxrwxr-x 2 insecureuser insecureuser 4096 Sep 23 16:34 evidencias
root@c1246db0b82f:~# █

```

Antes: ilustración 12

Endurecer la configuración de SSH

BACKUP y edito el archivo

ANTES: SECCION 3.5

Generar claves

```

proyecto@kali-117: ~
Archivo Acciones Editar Vista Ayuda
└─$ ssh-keygen -t ed25519 -f ~/.ssh/other20_key
Generating public/private ed25519 key pair.
Enter passphrase for "/home/proyecto/.ssh/other20_key" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/proyecto/.ssh/other20_key
Your public key has been saved in /home/proyecto/.ssh/other20_key.pub
The key fingerprint is:
SHA256:UkKqufy9CPDnyA2UnzUUT7k8F0ys9W3NHilwKTLFCTE proyecto@kali-117
The key's randomart image is:
+--[ED25519 256]--+
|      .+...E*00      |
|      O.*+ =,+      |
|      . 0x0+ * 0 .   |
|      * ..0+ . + =   |
|      = +.S . 0 .   |
|      = 0 0 .        |
|      * +           |
|      . 0 0         |
|      0 = 0.        |
+--[SHA256]-----+
proyecto@kali-117:~$ ssh-copy-id -i ~/.ssh/other20_key.pub root@192.117.30.20
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/proyecto/.ssh/other20_key.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
root@192.117.30.20's password:
Number of key(s) added: 1
Now try logging into the machine, with: "ssh -i /home/proyecto/.ssh/other20_key 'root@192.117.30.20'"
and check to make sure that only the key(s) you wanted were added.

```

Archivo configuración ssh

```
root@c124
Firefox ESR
Navegue por la web Ayuda
GNU nano 4.8 /etc/s
#ClientAliveInterval 0
#ClientAliveCountMax 3
#UseDNS no
#PidFile /var/run/sshd.pid
#MaxStartups 10:30:100
#PermitTunnel no
#ChrootDirectory none
#VersionAddendum none

# no default banner path
#Banner none

# Allow client to pass locale environment variables
AcceptEnv LANG LC_*

# override default of no subsystems
Subsystem      sftp      /usr/lib/openssh/sftp-server

# Example of overriding settings on a per-user basis
#Match User anoncvs
#      X11Forwarding no
#      AllowTcpForwarding no
#      PermitTTY no
#      ForceCommand cvs server
PermitEmptyPasswords no
PermitRootLogin no
PasswordAuthentication no
ClientAliveInterval 300
ClientAliveCountMax 2
MaxAuthTries 2
X11Forwarding no
Compression no
```

Verificar desde otra sesión

```
root@c1246db0b82f: ~
Archivo Acciones Editar Vista Ayuda
(proyecto@kali-117)-[~]
$ ssh -i ~/.ssh/other20_key root@192.117.30.20
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 6.14.0-37-generic x86_64)

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This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Last login: Fri Sep 25 17:35:08 2026 from 192.117.99.128
root@c1246db0b82f:~#
```

VERIFICAR CAMBIOS

```
Aplicaciones
Archivo Acciones Editar Vista Ayuda
root@c1246db0b82f: ~
$ ssh -i ~/.ssh/other20_key root@192.117.30.20
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 6.14.0-37-generic x86_64)

 * Documentation:  https://help.ubuntu.com
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This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Last login: Fri Sep 25 17:35:08 2026 from 192.117.99.128
root@c1246db0b82f:~# sshd -t
root@c1246db0b82f:~# systemctl restart sshd
root@c1246db0b82f:~# ssh -T | grep -Ei 'permitrootlogin|passwordauthentication|clientaliveinterval|maxauthtries|x11forwarding|compression'
usage: ssh [-46AaCfGgKkMnqsTtVvXxYy] [-B bind_interface]
          [-b bind_address] [-c cipher_spec] [-D [bind_address:]port]
          [-E log_file] [-e escape_char] [-F configfile] [-I pkcs11]
          [-i identity_file] [-J [user@]host[:port]] [-L address]
          [-l login_name] [-m mac_spec] [-O ctl_cmd] [-o option] [-p port]
          [-Q query_option] [-R address] [-S ctl_path] [-W host:port]
          [-w local_tun[:remote_tun]] destination [command]
root@c1246db0b82f:~# sshd -T | grep -Ei 'permitrootlogin|passwordauthentication|clientaliveinterval|maxauthtries|x11forwarding|compression'
maxauthtries 2
clientaliveinterval 300
permitrootlogin yes
passwordauthentication yes
x11forwarding yes
compression no
root@c1246db0b82f:~#
```

Configurar política de robustez y expiración de contraseñas

Instalar módulos

```
root@c1246db0b82f:~# apt update
Hit:1 http://archive.ubuntu.com/ubuntu focal InRelease
Get:2 http://archive.ubuntu.com/ubuntu focal-updates InRelease [128 kB]
Get:3 http://security.ubuntu.com/ubuntu focal-security InRelease [128 kB]
Get:4 http://archive.ubuntu.com/ubuntu focal-backports InRelease [128 kB]
Fetched 383 kB in 1s (345 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
2 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@c1246db0b82f:~# apt install libpam-pwquality -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  cracklib-runtime libcrack2 libpwquality-common libpwquality1 wamerican
The following NEW packages will be installed:
  cracklib-runtime libcrack2 libpam-pwquality libpwquality-common libpwquality1 wamerican
0 upgraded, 6 newly installed, 0 to remove and 2 not upgraded.
Need to get 410 kB of archives.
After this operation, 1939 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu focal/main amd64 libcrack2 amd64 2.9.6-3.2 [28.3 kB]
Get:2 http://archive.ubuntu.com/ubuntu focal/main amd64 cracklib-runtime amd64 2.9.6-3.2 [140 kB]
Get:3 http://archive.ubuntu.com/ubuntu focal/main amd64 libpwquality-common all 1.4.2-1build1 [7744 B]
Get:4 http://archive.ubuntu.com/ubuntu focal/main amd64 libpwquality1 amd64 1.4.2-1build1 [12.7 kB]
Get:5 http://archive.ubuntu.com/ubuntu focal/main amd64 libpam-pwquality amd64 1.4.2-1build1 [11.2 kB]
Get:6 http://archive.ubuntu.com/ubuntu focal/main amd64 wamerican all 2018.04.16-1 [210 kB]
Fetched 410 kB in 1s (565 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package libcrack2:amd64.
(Reading database ... 16893 files and directories currently installed.)
Preparing to unpack .../0-libcrack2_2.9.6-3.2_amd64.deb ...
Unpacking libcrack2:amd64 (2.9.6-3.2) ...
```

Config de common-password

```

GNU nano 4.8 /etc/pam.d/common-password
#
# /etc/pam.d/common-password - password-related modules common to all services
#
# This file is included from other service-specific PAM config files,
# and should contain a list of modules that define the services to be
# used to change user passwords. The default is pam_unix.
#
# Explanation of pam_unix options:
#
# The "sha512" option enables salted SHA512 passwords. Without this option,
# the default is Unix crypt. Prior releases used the option "md5".
#
# The "obscure" option replaces the old 'OBSCURE_CHECKS_ENAB' option in
# login.defs.
#
# See the pam_unix manpage for other options.
#
# As of pam 1.0.1-6, this file is managed by pam-auth-update by default.
# To take advantage of this, it is recommended that you configure any
# local modules either before or after the default block, and use
# pam-auth-update to manage selection of other modules. See
# pam-auth-update(8) for details.
#
# here are the per-package modules (the "Primary" block)
password      requisite          pam_pwquality.so retry=3 minlen=12 ucredit=-1 lcredit=-1 dcredit=-1
password      [success! default=ignore] pam_unix.so obscure use_authtok try_first_pass sha512
# here's the fallback if no module succeeds
password      requisite          pam_deny.so
# prime the stack with a positive return value if there isn't one already;
# this avoids us returning an error just because nothing sets a success code
# since the modules above will each just jump around
password      required           pam_permit.so

```

CONFIG EXPIRACION

```

GNU nano 4.8 /etc/login.defs
# mind.
#
# If USERGROUPS_ENAB is set to "yes", that will modify this UMASK default value
# for private user groups, i. e. the uid is the same as gid, and username is
# the same as the primary group name: for these, the user permissions will be
# used as group permissions, e. g. 022 will become 002.
#
# Prefix these values with "0" to get octal, "0x" to get hexadecimal.
#
# Password aging controls:
#
# PASS_MAX_DAYS      Maximum number of days a password may be used.
# PASS_MIN_DAYS      Minimum number of days allowed between password changes.
# PASS_WARN_AGE      Number of days warning given before a password expires.
#
# Min/max values for automatic uid selection in useradd
#
# UID_MIN             1000
# UID_MAX             60000
#
# System accounts
# SYS_UID_MIN         100
# SYS_UID_MAX         999

```

Aplicar la expiración a usuarios

```

root@c1246db0b82f:~# chage -M 90 -m 1 -W 7 insecureuser
root@c1246db0b82f:~# chage -l insecureuser
Last password change          : Jul 29, 2026
Password expires               : Oct 27, 2026
Password inactive              : never
Account expires                : never
Minimum number of days between password change : 1
Maximum number of days between password change : 90
Number of days of warning before password expires : 7

```

Consolidar http duplicado

Cambio

```

GNU nano 4.8
server {
    listen 80;
    #listen 8080;
    server_name _;

    root /var/www/html;
    autoindex on;
    client_max_body_size 100M;

    server_tokens on;

    location / {
        try_files $uri $uri/ =404;
    }
}

```

Guardar

```

root@c1246db0b82f:~# nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
root@c1246db0b82f:~# systemctl reload nginx

```

```

(proyecto@kali-117)-[~]
$ nmap -sV -p 8080 192.117.30.20
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-25 13:11 -05
Nmap scan report for 192.117.30.20.static.012.net.il (192.117.30.20)
Host is up (0.00074s latency).
PORT      STATE SERVICE      VERSION
8080/tcp  closed http-proxy
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.02 seconds

```

Activar reglas firewall

```

root@c1246db0b82f:~# ufw status verbose
Status: inactive
root@c1246db0b82f:~# ufw allow 22/tcp
Rules updated
root@c1246db0b82f:~# ufw allow 80/tcp
Rules updated
root@c1246db0b82f:~# ufw allow 25/tcp
Rules updated
root@c1246db0b82f:~# ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup

```

```

root@c1246db0b82f:~# ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), deny (routed)
New profiles: skip

To Action From
--
22/tcp ALLOW IN Anywhere
80/tcp ALLOW IN Anywhere
25/tcp ALLOW IN Anywhere

```

Mitigar el banner de info de postfix

```

root@c1246db0b82f:~# postconf smtpd_banner
smtpd_banner = $myhostname ESMTP $mail_name (Ubuntu)
root@c1246db0b82f:~# postconf -e 'smtpd_banner = $myhostname ESMTP'
root@c1246db0b82f:~# systemctl restart postfix
root@c1246db0b82f:~# postconf smtpd_banner
smtpd_banner = $myhostname ESMTP

```

Contraseña para modo de suario unico

```

root@c1246db0b82f:~# systemctl status emergency.service
● emergency.service - Emergency Shell
   Loaded: loaded (/lib/systemd/system/emergency.service; static; vendor preset: enabled)
   Active: inactive (dead)
   Docs: man:sulogin(8)
root@c1246db0b82f:~# systemctl status rescue.service
● rescue.service - Rescue Shell
   Loaded: loaded (/lib/systemd/system/rescue.service; static; vendor preset: enabled)
   Active: inactive (dead)
   Docs: man:sulogin(8)
root@c1246db0b82f:~# systemctl edit rescue.service
Editing "/etc/systemd/system/rescue.service.d/override.conf" canceled: temporary file is empty.
root@c1246db0b82f:~# systemctl edit emergency.service

```

```

[Service]
ExecStart=
ExecStart=-/sbin/sulogin

```

Scanner

```

rkhunter is already the newest version (1.4.6-8).
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
root@c1246db0b82f:~# rkhunter --update
Invalid WEB_CMD configuration option: Relative pathname: "/bin/false"

```

Crear perfiles de apparmor

```
Finished generating profile for /usr/sbin/postfix.  
root@c1246db0b82f:~# aa-enforce /etc/apparmor.d/usr.sbin.nginx  
Setting /etc/apparmor.d/usr.sbin.nginx to enforce mode.  
root@c1246db0b82f:~# aa-status  
apparmor module is loaded.  
173 profiles are loaded.  
92 profiles are in enforce mode.
```

No hay actualizaciones pendientes

APT-SHOW-VERSIONS

```

libruby2.7:amd64/focal-security 2.7.0-5ubuntu1.18 uptodate
libsasl2-2:amd64/focal-security 2.1.27+dfsg-2ubuntu0.1 uptodate
libsasl2-modules-db:amd64/focal-security 2.1.27+dfsg-2ubuntu0.1 uptodate
libseccomp2:amd64/focal-security 2.5.1-1ubuntu1~20.04.2 uptodate
libselinux1:amd64/focal 3.0-1build2 uptodate
libsemanage-common:all/focal 3.0-1build2 uptodate
libsemanage1:amd64/focal 3.0-1build2 uptodate
libsepol1:amd64/focal-security 3.0-1ubuntu0.1 uptodate
libsmartcols1:amd64/focal-security 2.34-0.1ubuntu9.6 uptodate
libsoup2.4-1:amd64/focal-security 2.70.0-1ubuntu0.5 uptodate
libsqlite3-0:amd64/focal-security 3.31.1-4ubuntu0.7 uptodate
libssl2:amd64/focal-updates 1.45.5-2ubuntu1.2 uptodate
libssl1.1:amd64/focal-security 1.1.1f-1ubuntu2.24 uptodate
libstdc++6:amd64/focal-security 10.5.0-1ubuntu1~20.04 uptodate
libstemmer0d:amd64/focal 0+svn585-2 uptodate
libsystemd0:amd64/focal-updates 245.4-4ubuntu3.24 uptodate
libtasn1-6:amd64/focal-security 4.16.0-2ubuntu0.1 uptodate
libtiff5:amd64/focal-security 4.1.0+git191117-2ubuntu0.20.04.14 uptodate
libtinfo6:amd64/focal-security 6.2-0ubuntu2.1 uptodate
libudev1:amd64/focal-updates 245.4-4ubuntu3.24 uptodate
libunistring2:amd64/focal 0.9.10-2 uptodate
libuuid1:amd64/focal-security 2.34-0.1ubuntu9.6 uptodate
libwebp6:amd64/focal-security 0.6.1-2ubuntu0.20.04.3 uptodate
libwrap0:amd64/focal 7.6.q-30 uptodate
libx11-6:amd64/focal-security 2:1.6.9-2ubuntu1.6 uptodate
libx11-data:amd64/focal-security 2:1.6.9-2ubuntu1.6 uptodate
libxau6:amd64/focal 1:1.0.9-0ubuntu1 uptodate
libxcb1:amd64/focal 1.14-2 uptodate
libxdmcp6:amd64/focal 1:1.1.3-0ubuntu1 uptodate
libxml2:amd64/focal-security 2.9.10+dfsg-5ubuntu0.20.04.10 uptodate
libxpm4:amd64/focal-security 1:3.5.12-1ubuntu0.20.04.2 uptodate
libxslt1.1:amd64/focal-security 1.1.34-4ubuntu0.20.04.3 uptodate
libxtables12:amd64/focal-updates 1.8.4-3ubuntu2.1 uptodate
libyaml-0-2:amd64/focal 0.2.2-1 uptodate
libzstd1:amd64/focal-security 1.4.4+dfsg-3ubuntu0.1 uptodate
login:amd64/focal-security 1:4.8.1-1ubuntu5.20.04.5 uptodate
logsave:amd64/focal-updates 1.45.5-2ubuntu1.2 uptodate
lsb-base:all/focal 11.1.0ubuntu2 uptodate
lsb-release:all/focal 11.1.0ubuntu2 uptodate
lsyf:amd64/focal-updates 4.93.2+dfsg-1ubuntu0.20.04.1 uptodate
lynis:all/focal 2.6.2-1 uptodate
mawk:amd64/focal 1.3.4.20200120-2 uptodate
mime-support:all/focal 3.64ubuntu1 uptodate
mount:amd64/focal-security 2.34-0.1ubuntu9.6 uptodate
nano:amd64/focal-security 4.8-1ubuntu1.1 uptodate
ncurses-base:all/focal-security 6.2-0ubuntu2.1 uptodate
ncurses-bin:amd64/focal-security 6.2-0ubuntu2.1 uptodate
net-tools:amd64/focal-security 1.60+git20180626.aebd88e-1ubuntu1.3 uptodate
netbase:all/focal 6.1 uptodate
nginx:all/focal-security 1.18.0-0ubuntu1.7 uptodate
nginx-common:all/focal-security 1.18.0-0ubuntu1.7 uptodate
nginx-core:amd64/focal-security 1.18.0-0ubuntu1.7 uptodate
openssh-client:amd64/focal-security 1:8.2p1-4ubuntu0.13 uptodate
openssh-server:amd64/focal-security 1:8.2p1-4ubuntu0.13 uptodate
openssh-sftp-server:amd64/focal-security 1:8.2p1-4ubuntu0.13 uptodate
openssl:amd64/focal-security 1.1.1f-1ubuntu2.24 uptodate
packagekit:amd64/focal-security 1.1.13-2ubuntu1.1 uptodate
passwd:amd64/focal-security 1:4.8.1-1ubuntu5.20.04.5 uptodate
perl:amd64/focal-security 5.30.0-9ubuntu0.5 uptodate
perl-base:amd64/focal-security 5.30.0-9ubuntu0.5 uptodate
perl-modules-5.30:all/focal-security 5.30.0-9ubuntu0.5 uptodate

```

Archivo Acciones

projectbook

apt-show

No se ha encontrado

sudo apt install

requiere instalar

sudo apt install

[sudo] contrase

Installing:

projectbook

Summary:

Upgrading: 0;

Download size:

Space needed:

Error: http://ht

404 Not Found

[Err] Fallo al

[230 80]

[Err] No se pu

projectbook

apt-show

No se ha encontrado

sudo apt install

requiere instala

projectbook

apt-show

No se ha encontrado

sudo apt install

requiere instala

projectbook

apt-show

No se ha encontrado

sudo apt install

requiere instala

projectbook

apt-show

No se ha encontrado

sudo apt install

requiere instala

projectbook

apt-show

No se ha encontrado

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No se ha encontrado

sudo apt install

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projectbook

apt-show

No se ha encontrado

sudo apt install

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projectbook

apt-show